

15.7 THE DIVERGENCE THEOREM

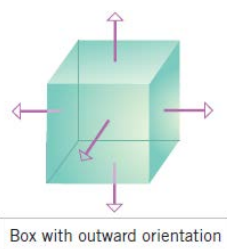
In earlier section, we defined the divergence of a vector field

$$\mathbf{F}(x, y, z) = f(x, y, z)\mathbf{i} + g(x, y, z)\mathbf{j} + h(x, y, z)\mathbf{k}$$

as

$$\operatorname{div} \mathbf{F} = \frac{\partial f}{\partial x} + \frac{\partial g}{\partial y} + \frac{\partial h}{\partial z}$$

but we did not attempt to give a physical explanation of its meaning at that time. The following result, known as the **Divergence Theorem** or **Gauss's Theorem**, will provide us with a physical interpretation of divergence in the context of fluid flow.



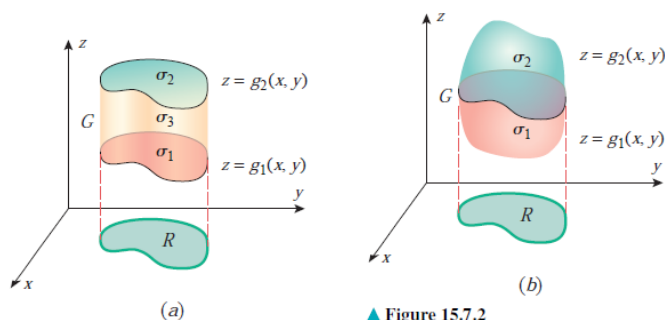
▲ Figure 15.7.1

15.7.1 THEOREM (The Divergence Theorem) Let G be a solid whose surface σ is oriented outward. If

$$\mathbf{F}(x, y, z) = f(x, y, z)\mathbf{i} + g(x, y, z)\mathbf{j} + h(x, y, z)\mathbf{k}$$

where f , g , and h have continuous first partial derivatives on some open set containing G , and if $\mathbf{n}$ is the outward unit normal on σ , then

$$\iint_{\sigma} \mathbf{F} \cdot \mathbf{n} \, dS = \iiint_G \operatorname{div} \mathbf{F} \, dV \quad (1)$$



▲ Figure 15.7.2

PROOF Formula (1) can be expressed as

$$\iint_{\sigma} [f(x, y, z)\mathbf{i} + g(x, y, z)\mathbf{j} + h(x, y, z)\mathbf{k}] \cdot \mathbf{n} dS = \iiint_G \left(\frac{\partial f}{\partial x} + \frac{\partial g}{\partial y} + \frac{\partial h}{\partial z} \right) dV$$

so it suffices to prove the three equalities

$$\iint_{\sigma} f(x, y, z)\mathbf{i} \cdot \mathbf{n} dS = \iiint_G \frac{\partial f}{\partial x} dV \quad (2a)$$

$$\iint_{\sigma} g(x, y, z)\mathbf{j} \cdot \mathbf{n} dS = \iiint_G \frac{\partial g}{\partial y} dV \quad (2b)$$

$$\iint_{\sigma} h(x, y, z)\mathbf{k} \cdot \mathbf{n} dS = \iiint_G \frac{\partial h}{\partial z} dV \quad (2c)$$

Since the proofs of all three equalities are similar, we will prove only the third.

Suppose that G has upper surface $z = g_2(x, y)$, lower surface $z = g_1(x, y)$, and projection R on the xy -plane. Let σ_1 denote the lower surface, σ_2 the upper surface, and σ_3 the lateral surface (Figure 15.7.2a). If the upper surface and lower surface meet as in

Figure 15.7.2b, then there is no lateral surface σ_3 . Our proof will allow for both cases shown in those figures.

It follows from Theorem 14.5.2 that

$$\iiint_G \frac{\partial h}{\partial z} dV = \iint_R \left[\int_{g_1(x, y)}^{g_2(x, y)} \frac{\partial h}{\partial z} dz \right] dA = \iint_R \left[h(x, y, z) \right]_{z=g_1(x, y)}^{g_2(x, y)} dA$$

so

$$\iiint_G \frac{\partial h}{\partial z} dV = \iint_R [h(x, y, g_2(x, y)) - h(x, y, g_1(x, y))] dA \quad (3)$$

Next we will evaluate the surface integral in (2c) by integrating over each surface of G separately. If there is a lateral surface σ_3 , then at each point of this surface $\mathbf{k} \cdot \mathbf{n} = 0$ since $\mathbf{n}$ is horizontal and $\mathbf{k}$ is vertical. Thus,

$$\iint_{\sigma_3} h(x, y, z) \mathbf{k} \cdot \mathbf{n} dS = 0$$

Therefore, regardless of whether G has a lateral surface, we can write

$$\iint_{\sigma} h(x, y, z) \mathbf{k} \cdot \mathbf{n} dS = \iint_{\sigma_1} h(x, y, z) \mathbf{k} \cdot \mathbf{n} dS + \iint_{\sigma_2} h(x, y, z) \mathbf{k} \cdot \mathbf{n} dS \quad (4)$$

On the upper surface σ_2 , the outer normal is an upward normal, and on the lower surface σ_1 , the outer normal is a downward normal. Thus, Formulas (12) and (13) of Section 15.6 imply that

$$\begin{aligned} \iint_{\sigma_2} h(x, y, z) \mathbf{k} \cdot \mathbf{n} dS &= \iint_R h(x, y, g_2(x, y)) \mathbf{k} \cdot \left(-\frac{\partial z}{\partial x} \mathbf{i} - \frac{\partial z}{\partial y} \mathbf{j} + \mathbf{k} \right) dA \\ &= \iint_R h(x, y, g_2(x, y)) dA \end{aligned} \quad (5)$$

and

$$\begin{aligned} \iint_{\sigma_1} h(x, y, z) \mathbf{k} \cdot \mathbf{n} dS &= \iint_R h(x, y, g_1(x, y)) \mathbf{k} \cdot \left(\frac{\partial z}{\partial x} \mathbf{i} + \frac{\partial z}{\partial y} \mathbf{j} - \mathbf{k} \right) dA \\ &= - \iint_R h(x, y, g_1(x, y)) dA \end{aligned} \quad (6)$$

Substituting (5) and (6) into (4) and combining the terms into a single integral yields

$$\iint_{\sigma} h(x, y, z) \mathbf{k} \cdot \mathbf{n} dS = \iint_R [h(x, y, g_2(x, y)) - h(x, y, g_1(x, y))] dA \quad (7)$$

Equation (2c) now follows from (3) and (7). ■

The flux of a vector field across a closed surface with outward orientation is sometimes called the **outward flux** across the surface. In words, the Divergence Theorem states:

The outward flux of a vector field across a closed surface is equal to the triple integral of the divergence over the region enclosed by the surface.

■ USING THE DIVERGENCE THEOREM TO FIND FLUX

► **Example 1** Use the Divergence Theorem to find the outward flux of the vector field $\mathbf{F}(x, y, z) = z\mathbf{k}$ across the sphere $x^2 + y^2 + z^2 = a^2$.

Solution. Let σ denote the outward-oriented spherical surface and G the region that it encloses. The divergence of the vector field is

$$\operatorname{div} \mathbf{F} = \frac{\partial z}{\partial z} = 1$$

so from (1) the flux across σ is

$$\Phi = \iint_{\sigma} \mathbf{F} \cdot \mathbf{n} \, dS = \iiint_G dV = \text{volume of } G = \frac{4\pi a^3}{3}$$

Remark:

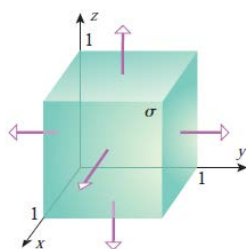
Note how much simpler this calculation is than that in Example 1 of Section 15.6.

The Divergence Theorem is usually the method of choice for finding the flux across closed piecewise smooth surfaces with multiple sections, since it eliminates the need for a separate integral evaluation over each section. This is illustrated in the next three examples.

► **Example 2** Use the Divergence Theorem to find the outward flux of the vector field

$$\mathbf{F}(x, y, z) = 2x\mathbf{i} + 3y\mathbf{j} + z^2\mathbf{k}$$

across the unit cube in Figure 15.7.3.



▲ Figure 15.7.3

Solution. Let σ denote the outward-oriented surface of the cube and G the region that it encloses. The divergence of the vector field is

$$\operatorname{div} \mathbf{F} = \frac{\partial}{\partial x}(2x) + \frac{\partial}{\partial y}(3y) + \frac{\partial}{\partial z}(z^2) = 5 + 2z$$

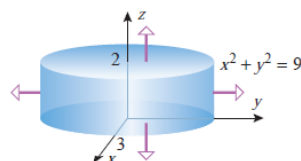
so from (1) the flux across σ is

$$\begin{aligned} \Phi &= \iint_{\sigma} \mathbf{F} \cdot \mathbf{n} \, dS = \iiint_G (5 + 2z) \, dV = \int_0^1 \int_0^1 \int_0^1 (5 + 2z) \, dz \, dy \, dx \\ &= \int_0^1 \int_0^1 [5z + z^2]_{z=0}^1 \, dy \, dx = \int_0^1 \int_0^1 6 \, dy \, dx = 6 \quad \blacktriangleleft \end{aligned}$$

► **Example 3** Use the Divergence Theorem to find the outward flux of the vector field

$$\mathbf{F}(x, y, z) = x^3 \mathbf{i} + y^3 \mathbf{j} + z^2 \mathbf{k}$$

across the surface of the region that is enclosed by the circular cylinder $x^2 + y^2 = 9$ and the planes $z = 0$ and $z = 2$ (Figure 15.7.4).



▲ Figure 15.7.4

Solution. Let σ denote the outward-oriented surface and G the region that it encloses. The divergence of the vector field is

$$\operatorname{div} \mathbf{F} = \frac{\partial}{\partial x}(x^3) + \frac{\partial}{\partial y}(y^3) + \frac{\partial}{\partial z}(z^2) = 3x^2 + 3y^2 + 2z$$

so from (1) the flux across σ is

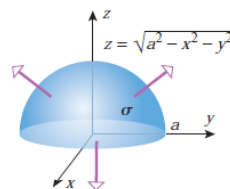
$$\begin{aligned}
 \Phi &= \iint_{\sigma} \mathbf{F} \cdot \mathbf{n} \, dS = \iiint_G (3x^2 + 3y^2 + 2z) \, dV \\
 &= \int_0^{2\pi} \int_0^3 \int_0^2 (3r^2 + 2z)r \, dz \, dr \, d\theta \\
 &= \int_0^{2\pi} \int_0^3 [3r^3z + z^2r]_{z=0}^2 \, dr \, d\theta \\
 &= \int_0^{2\pi} \int_0^3 (6r^3 + 4r) \, dr \, d\theta \\
 &= \int_0^{2\pi} \left[\frac{3r^4}{2} + 2r^2 \right]_0^3 \, d\theta \\
 &= \int_0^{2\pi} \frac{279}{2} \, d\theta = 279\pi \quad \blacktriangleleft
 \end{aligned}$$

Using cylindrical coordinates

► **Example 4** Use the Divergence Theorem to find the outward flux of the vector field

$$\mathbf{F}(x, y, z) = x^3\mathbf{i} + y^3\mathbf{j} + z^3\mathbf{k}$$

across the surface of the region that is enclosed by the hemisphere $z = \sqrt{a^2 - x^2 - y^2}$ and the plane $z = 0$ (Figure 15.7.5).



▲ Figure 15.7.5

Solution. Let σ denote the outward-oriented surface and G the region that it encloses. The divergence of the vector field is

$$\operatorname{div} \mathbf{F} = \frac{\partial}{\partial x}(x^3) + \frac{\partial}{\partial y}(y^3) + \frac{\partial}{\partial z}(z^3) = 3x^2 + 3y^2 + 3z^2$$

so from (1) the flux across σ is

$$\begin{aligned}\Phi &= \iint_{\sigma} \mathbf{F} \cdot \mathbf{n} \, dS = \iiint_G (3x^2 + 3y^2 + 3z^2) \, dV \\&= \int_0^{2\pi} \int_0^{\pi/2} \int_0^a (3\rho^2) \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta \\&= 3 \int_0^{2\pi} \int_0^{\pi/2} \int_0^a \rho^4 \sin \phi \, d\rho \, d\phi \, d\theta \\&= 3 \int_0^{2\pi} \int_0^{\pi/2} \left[\frac{\rho^5}{5} \sin \phi \right]_{\rho=0}^a \, d\phi \, d\theta \\&= \frac{3a^5}{5} \int_0^{2\pi} \int_0^{\pi/2} \sin \phi \, d\phi \, d\theta \\&= \frac{3a^5}{5} \int_0^{2\pi} [-\cos \phi]_0^{\pi/2} \, d\theta \\&= \frac{3a^5}{5} \int_0^{2\pi} d\theta = \frac{6\pi a^5}{5} \blacktriangleleft\end{aligned}$$

Using spherical
coordinates

Practice Problem:

Chapter-15.7:

Problems: 9--16.

Thank you for your attention!